Weight and local gravity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish mass from gravitational weight.

SUCCESS CRITERIA - I can...

1. Use mass in kg and force in N.
2. Calculate weight using the stated local g.
3. Explain how weight changes while mass stays the same.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A simplified model predicts position (-50, 70) km with bounds ±5 km in x and ±7 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$; $g = F_g/m$.
F_g is weight. Field strength in N/kg is numerically equal to free-fall acceleration in m/s².

Worked reasoning: A 3 kg object in a field of 4 N/kg has weight 3×4=12 N. Moving it to a 2 N/kg field halves weight to 6 N but leaves mass 3 kg.

Your turn: Draw weight arrows for the two places and label the unchanged quantity.

Location	Mass (kg)	g (N/kg)	Weight (N)
A	3	4	12
B	3	2	6

Marzano check - circle: ① ② ③

03 Read and interpret the model

The same 2 kg object visits worlds with g=10 and g=5 N/kg. Compare its mass and weight.

World	g (N/kg)
A	10
B	5

Marzano check - circle: ① ② ③

04 Apply the physics

An astronaut has mass 60 kg. Complete the location table using W = mg. Explain which property of the astronaut stays the same.

Location	g (N/kg)	Weight (N)
Earth	9.8	____
Moon	1.6	____
Planet	____	240

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A scale reports 0 N for an astronaut in an orbiting spacecraft. Does that prove Earth exerts no gravitational force? Explain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg cart has 6 N right and 38 N left. Find acceleration. What else must you know to decide if it speeds up? Draw the forces on the chosen object and label their sources.



Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

An object weighs 49 N where g=9.8 N/kg. Find its mass and weight where g=2 N/kg.



Add and label force arrows.

Draw and label the gravitational force.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An object weighs 72 N at one location and 18 N at another where g=3 N/kg. Find its mass and the first field strength.

Case	Mass (kg)	g (N/kg)	Weight (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3